

Name: _____

Date: _____



Year 10 General Mathematics

Test 2, 2015

Topics – Surds, Indices, Algebra

Total Time: **60** minutes

Weighting: 5% of the year.

Equipment Allowed: $\frac{1}{2}$ page notes (A4 one side)

60
= _____ %

CALCULATOR FREE TEST

1. [2 marks: 1, 1]

Write these surds in simplest form:

a) $\sqrt{28}$
 $\sqrt{4 \times 7}$
 $2\sqrt{7}$

b) $\sqrt{700}$
 $\sqrt{100 \times 7}$
 $10\sqrt{7}$

2. [6 marks: 1,2,2,3]

Simplify each of the following:

a) $2\sqrt{3} + 3\sqrt{3} - \sqrt{3}$
 $\sqrt{3}(2+3-1)$
 $= 4\sqrt{3}$

b) $8\sqrt{7} + 5\sqrt{13} + 3\sqrt{7} - 2\sqrt{13}$
 $\sqrt{7}(8+3) + \sqrt{13}(5-2)$
 $11\sqrt{7} + 3\sqrt{13}$

c) $\sqrt{32} + 3\sqrt{2}$
 $\sqrt{16 \times 2} + 3\sqrt{2}$
 $4\sqrt{2} + 3\sqrt{2}$
 $= 7\sqrt{2}$

d) $2\sqrt{12} + 7\sqrt{3} - \sqrt{75}$
 $2\sqrt{3 \times 4} + 7\sqrt{3} - \sqrt{25 \times 3}$
 $4\sqrt{3} + 7\sqrt{3} - 5\sqrt{3}$
 $= 6\sqrt{3}$

3. 3 [marks: 1, 2]

Rationalise the denominator:

a) $\frac{1}{\sqrt{11}} \times \frac{\sqrt{11}}{\sqrt{11}} = \frac{\sqrt{11}}{11}$

b) $\frac{4\sqrt{5}}{3\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}} = \frac{4\sqrt{35}}{21}$

4. [4 marks: 1, 1, 1, 1]

Use index laws to simplify the following, leaving your answers with positive indices.

a) $6^8 \times 6^{15}$
 $6^{(8+15)}$
 $= 6^{23}$

b) $5^{-5} \times 5^8$
 5^{-5+8}
 $= 5^3$

c) $(4^{-2})^5$
 $4^{-2 \times 5}$
 $4^{-10} = \frac{1}{4^{10}}$

d) $7^8 \times 7^{-3} \times 7^0$
 7^{8-3+0}
 $= 7^5$

5. [9 marks: 2, 2, 2, 3]

Simplify the following, leaving your answers in index form.

a) $(6^2 \times 5^{-2})^3$
 $6^{2 \times 3} \times 5^{-2 \times 3} = 6^6 \times 5^{-6}$
 $= \frac{6^6}{5^6} = \left(\frac{6}{5}\right)^6$

b) $9^3 \times 3^{-3} \times 27^{-2}$
 $3^6 \times 3^{-3} \times (3^3)^{-2}$
 $= 3^6 \times 3^{-3} \times 3^{-6}$
 $= \frac{1}{3^3} \text{ or } \frac{1}{27} \times 3^{-3}$

c) $\frac{8^3 \times 3^2}{(3^5 \times 2^2)^3}$
 $= \frac{(2^3)^3 \times 3^2}{3^{15} \times 2^6} = \frac{2^9 \times 3^2}{3^{15} \times 2^6}$
 $= 2^3 \times 3^{-13}$

d) $\frac{4^3 \times 3^{15}}{(3^5 \times 2^2)^4}$
 $= \frac{(2^2)^3 \times 3^{15}}{3^{20} \times 2^8} = \frac{2^6 \times 3^{15}}{3^{20} \times 2^8}$
 $= 2^{-2} \times 3^{-5}$

6. [12 marks: 1, 2, 2, 2, 2, 3]

Expand and where possible, simplify the following

a) $4(x + 3)$

$4x + 12$

b) $2a(a + 4)$

$2a^2 + 8a$

c) $-5b(3b - 8)$

$-15b^2 + 40b$

or $40b - 15b^2$

d) $(g + 3)^2$

$= (g + 3)(g + 3)$

$= g^2 + 3g + 3g + 9$

$= g^2 + 6g + 9$

e) $(x + 5)(x - 3)$

$x^2 - 3x + 5x - 15$

$x^2 + 2x - 15$

d) $(x + 5)(x - 3)(2x + 1)$

$(x^2 + 2x - 15)(2x + 1)$

$= 2x^3 + 4x^2 - 30x + x^2 + 2x - 15$

$= 2x^3 + 5x^2 - 28x - 15$

7. [24 marks: 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3]

Factorise the following

a) $28y - 42$

$7(4y - 6)$

b) $6x + 18$

$3(2x + 6)$ 1 mark

$6(x + 3)$ 2 marks

c) $x^2 + 4x$

$x(x + 4)$

d) $3z^2 - 9z$

$3z(z - 3)$

e) $2(q+4)+p(q+4)$

$$(2+p)(q+4)$$

f) $x^2 + 4x + 3$

$$(x+3)(x+1)$$

c) $25 - 16y^2$

$$(5+4y)(5-4y)$$

d) $(a+8)^2 - 49$

$$(a+8-7)(a+8+7)$$

$$= (a+1)(a+15)$$

e) $x^2 + x - 12$

$$(x-3)(x+4)$$

f) $4x^2 + 36x + 80$

$$= 4(x^2 + 9x + 20)$$

$$= 4(x+4)(x+5)$$

or similar

g) $3x^2 - 13x - 10$

$$(3x+2)(x-5)$$

End Of Test